

DHARUN KANAGASABAI

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PROFESSIONAL SUMMARY

Data Analyst with an MSc in Data Science (University of Sheffield) and three years of enterprise data delivery experience at EY, applying statistical modelling, machine learning, and SQL/Python-based analysis to large, complex datasets. Built a five-agent AI pipeline processing 81,864 archival records, independently designing a precision/recall/F1 evaluation framework that lifted extraction quality from 63% to 94% F1 across three debugging rounds. Applied survival analysis and probabilistic sensitivity modelling to oncology clinical data at Syneos Health, and delivered Oracle Fusion (HCM, FSCM, EBS) reporting and integration solutions across enterprise finance and HR systems. Comfortable owning analytical problems end-to-end, from requirements through to validated, stakeholder-ready outputs, and communicating technical findings clearly to non-technical audiences.

TECHNICAL SKILLS

Programming & Statistics: Python (pandas, scikit-learn, NumPy, SciPy, matplotlib, seaborn, statsmodels), R (statistical modelling, Shiny), SQL, PL/SQL, regression, survival analysis (Generalised Gamma, Gompertz, fractional polynomial), K-means clustering, classification (XGBoost, logistic regression), probabilistic sensitivity analysis, precision/recall/F1 evaluation frameworks

Data Engineering & Cloud: ETL/ELT pipeline design, REST API integration, dbt, Terraform, Docker, GitHub Actions/CI-CD, Google Cloud Platform (BigQuery, Cloud Run, GCS), Microsoft Azure/Databricks (PySpark), Neo4j

Oracle & Enterprise Systems: Oracle Fusion Cloud (HCM, FSCM, EBS), Oracle Integration Cloud (OIC), Oracle APEX, VBCS, BI Publisher, OTBI, Fast Formula, business analysis, requirements gathering, functional design documentation

BI & Visualisation: Power BI, R Shiny (interactive dashboards), advanced Microsoft Excel (Pivot Tables, automation logic)

Data Quality & Documentation: data validation, reconciliation, data profiling and mapping, dimensional modelling (star/snowflake schema), technical documentation, audit-readiness

Collaboration & Communication: cross-functional stakeholder engagement, technical writing and presentation, translating analytical findings for non-technical audiences

PROFESSIONAL EXPERIENCE

Health Economics & Outcomes Research (HEOR) Intern | *Syneos Health, UK* | *Mar 2026 – Sep 2026*

- Delivered a production R Shiny cost-effectiveness analysis application replicating an NHS oncology health-economic model end-to-end, validated to within 0.0001% of the reference Excel model, ensuring full reproducibility and audit-readiness.
- Selected and validated a time-to-event model for the oncology cost-effectiveness case by applying and comparing survival analysis techniques (Generalised Gamma, Gompertz, fractional polynomial), then built a probabilistic sensitivity analysis (PSA) module running 500 iterations in seconds, versus several minutes per run in the reference Excel model, to stress-test outputs across parameter uncertainty.
- Wrote Python and SQL scripts for data extraction, transformation, and quality validation, reconciling 7 named clinical and real-world data sources (AENEAS, FLAURA, Butcher ITC/MAIC/STC, NHS Reference Costs 2019/20, PSSRU 2021, NICE TA654, BNF 2021) into a single structured application consolidating a 19-sheet, 63,000+ formula reference Excel model.
- Built a scenario save/load system producing styled .xlsx outputs and a comparator-upload module for external data ingestion, replacing manual cell-by-cell navigation and recalculation with a single slider-and-run interaction for common scenario changes (e.g., updating a PAS discount); presented the model, outputs, and limitations to director-level stakeholders via a 19-slide technical deck.
- Expanded and maintained an internal chatbot knowledge base to support model users across the analytics team.

Technology Consultant (Senior Analyst → Associate Consultant → Consultant) | *Ernst & Young (EY), India* | *2022 – 2025*

- Promoted three times in three years and received two internal EY awards for delivery excellence: 'Collaboration Extraordinaire' (2025) and 'I Am Exceptional' (2023).
- Led delivery of Oracle Fusion reporting and integration workstreams (BI Publisher, OTBI, OIC, HCM Extracts) across 5 enterprise engagements (3 implementations, 2 support projects) for Finance and HR

clients, owning data quality and validation across DEV, TEST, and PROD environments to ensure consistency and accuracy of reporting outputs.

- Designed and deployed Oracle OIC integrations, BI Publisher reports, VBCS applications, APEX solutions, and HCM Extracts/Fast Formulas across the Oracle Fusion and EBS estate, translating ambiguous business requirements into working, tested systems under live client deadlines.
- Wrote complex SQL and PL/SQL queries and built dimensional data models (star/snowflake schema) for data extraction, transformation, and reporting, optimising query performance across large enterprise datasets.
- Engineered Python and React JS proof-of-concept tools to demonstrate analytical capability on enterprise data sources and acted as technical advisor to vendors on Oracle database and integration issues.

PROJECTS

Agentic AI Pipeline for Knowledge Graph Generation from Archival Data | *MSc Dissertation, University of Sheffield* | 2025 – 2026

- Improved entity and relation extraction quality from 62.7% to 94.0% F1 across three rounds of root-cause debugging, by independently designing a hand-labelled 20-record gold evaluation set and a precision/recall/F1 scoring framework.
- Built a five-agent Python pipeline (document processing, entity extraction, relation extraction, ontology alignment, knowledge graph construction) coordinated by a shared-memory orchestrator, processing 81,864 records from a live archive API (National Fairground and Circus Archive, CONTENTdm).
- Integrated the Anthropic API for LLM-based entity/relation extraction as a swappable component benchmarked against a rule-based baseline (spaCy, rapidfuzz), aligned outputs to the CIDOC-CRM cultural-heritage ontology, and exported the resulting knowledge graph to GraphML, JSON, RDF/Turtle, and Neo4j-ready Cypher.

Real-Time Market Data Streaming Pipeline | *Independent Project* | 2026

- Built and deployed an end-to-end streaming pipeline, a Redpanda/Kafka producer ingesting a live exchange WebSocket feed, a stateful consumer computing rolling VWAP and OHLC candles into Redis, and a FastAPI layer with API-key auth and Redis-backed tiered rate limiting — to GCP Compute Engine and Cloud Run, orchestrated locally with Docker Compose and deployed via GitHub Actions CI.
- Wrote 13 automated tests covering aggregation logic, authentication, and rate limiting against fakeredis, validating the system with no live infrastructure required.

Cloud-Native ELT Pipeline | *Independent Project* | 2026

- Built a Python extractor landing raw NDJSON in Google Cloud Storage with a BigQuery external table reading the files directly, then modelled staging, intermediate, and mart layers in dbt with schema tests, mirroring medallion-architecture principles.
- Provisioned all infrastructure (15 resources) via Terraform using Workload Identity Federation instead of static service-account keys, running the full pipeline through GitHub Actions.

Player Churn & Engagement Propensity Modelling | *Independent Project* | 2026

- Built an end-to-end churn prediction pipeline on publicly available gaming engagement data, training XGBoost and logistic regression classifiers on engineered behavioural features (session frequency, recency, play duration, monetisation patterns).
- Segmented the player base via K-means clustering on behavioural features into distinct engagement personas with different churn-risk profiles, tuning classification thresholds for a retention-targeting use case.

EDUCATION

MSc Data Science | *University of Sheffield, UK* | Sep 2025 – Sep 2026

Core modules: Machine Learning, Statistical Modelling, Bayesian Methods, Data Visualisation, Big Data Technologies (Azure Databricks, PySpark), Data & Society

BTech, Mechanical Engineering | *Dayananda Sagar University, India* | Sep 2018 – Sep 2022

AWARDS

- 'Collaboration Extraordinaire' — EY Awards, 2025
- 'I Am Exceptional' — EY Awards, 2023